

Manufacturing Process Executive Summary

Key Performance Indicators

Simulation LOTs: 8,000

Normal LOTs: 6,973 (87.16%)

Defect LOTs: 1,027 (12.84%)

Overall Average Yield: 97.45%

Normal LOT Average Yield: 98.60%

Defect Breakdown

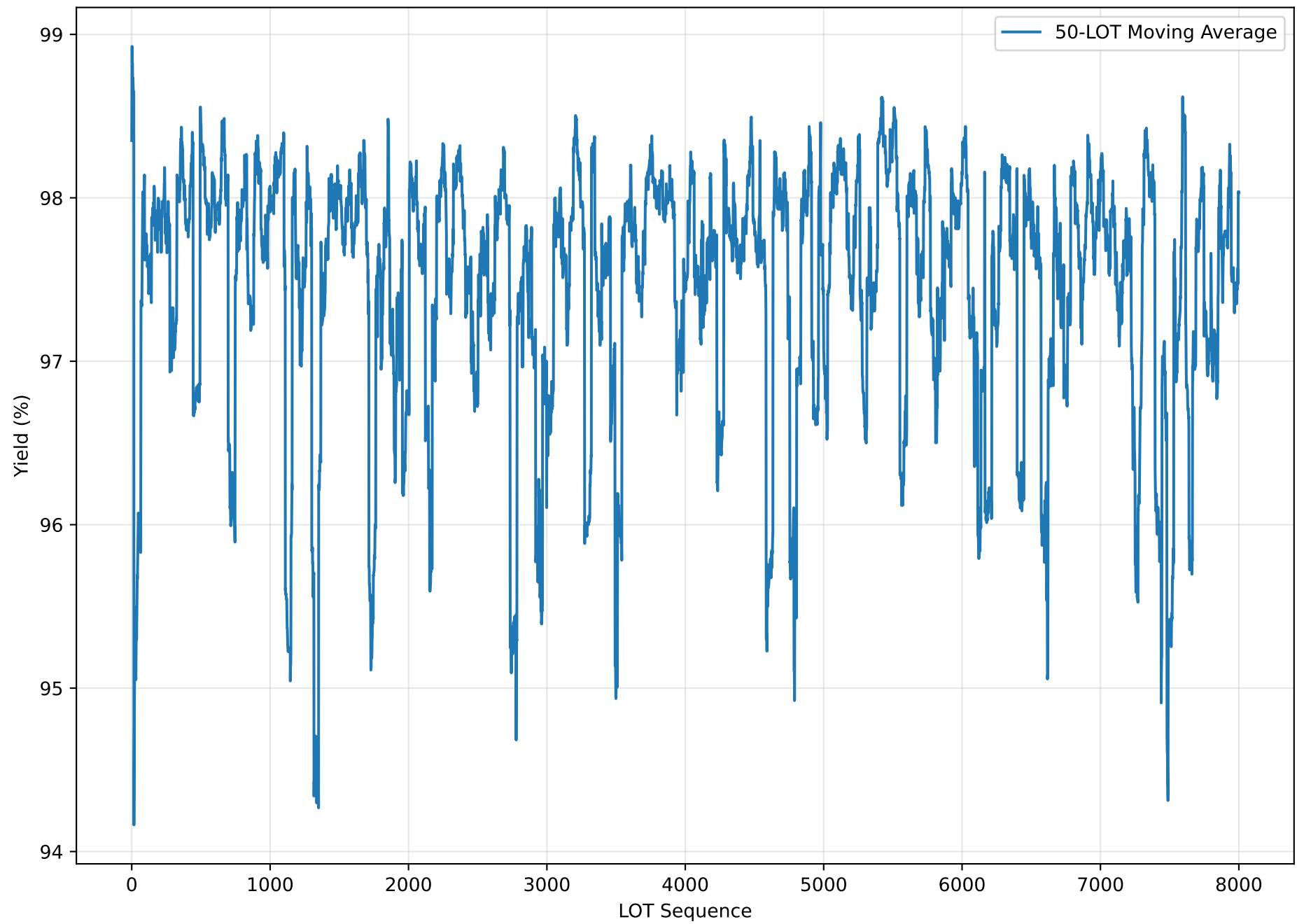
- Delamination: 1,027 LOTs (12.84%)

Conclusion

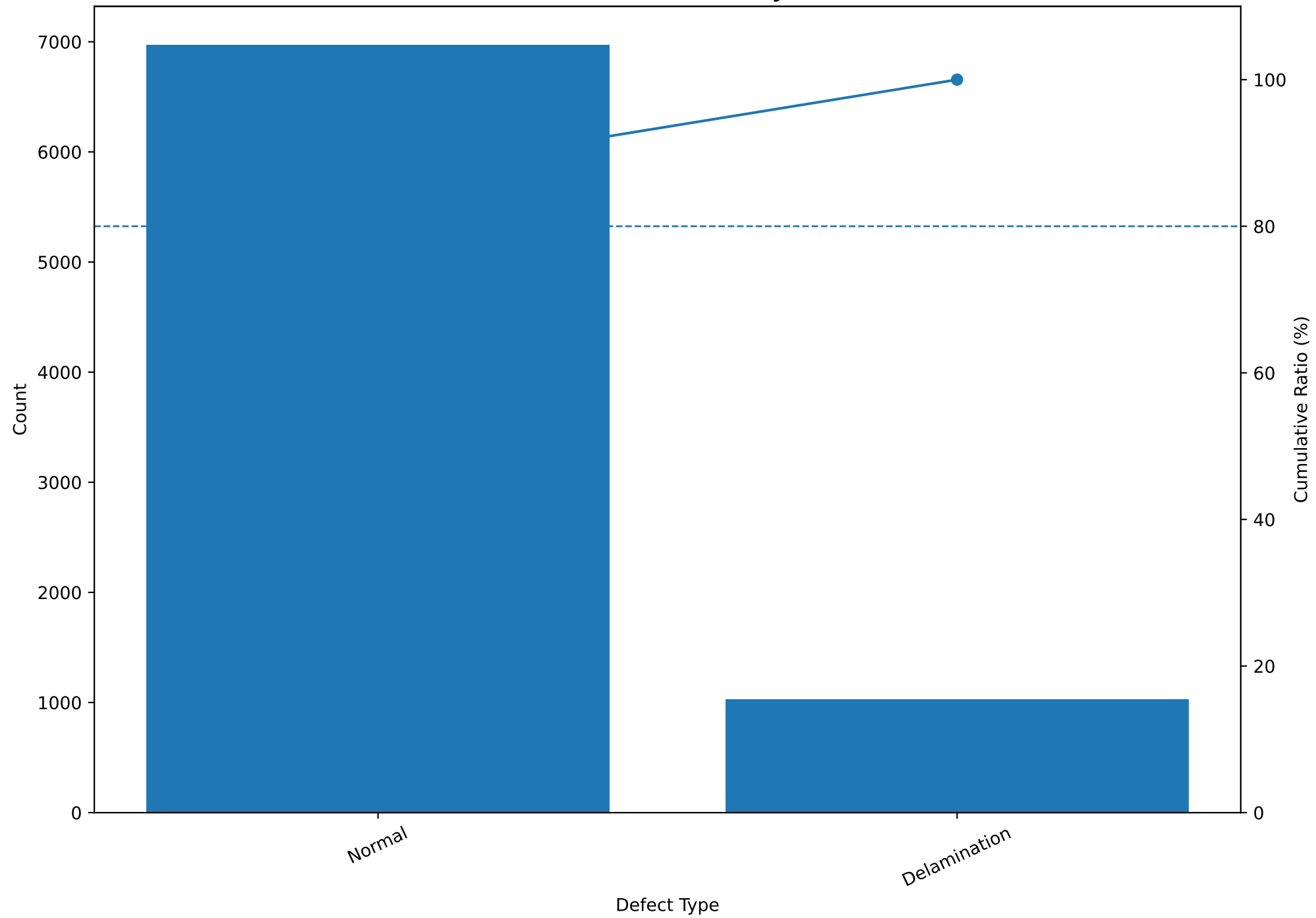
Based on the current 8,000-LOT simulation, the expected overall yield is 97.45% and the expected defect rate is 12.84%. The largest defect category is Delamination, 1,027 LOTs.

Note: The values above are simulation results generated from the current process model, not actual production forecasts.

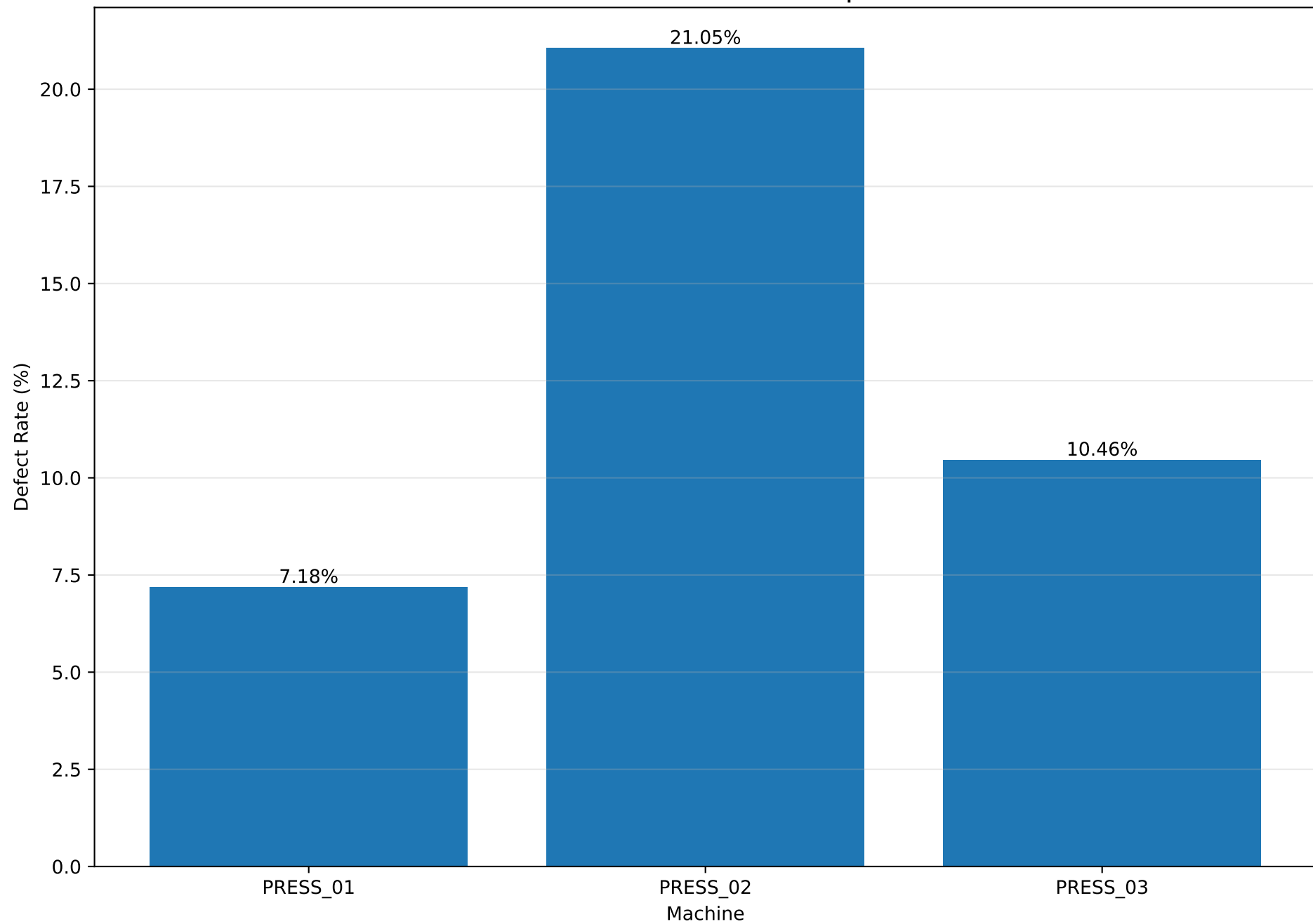
Yield Trend



Defect Pareto Analysis



Machine Defect Rate Comparison



Process Capability Summary

Parameter	Mean	Std	LSL	USL	Index	Value	Assessment
CZ_Roughness	349.931	8.225	320.0	380.0	Cpk	1.213	Marginal
Total_Thickness	32.025	0.262	30.0	34.0	Cpk	2.517	Excellent
Peel_Strength	496.755	22.33	400.0	nan	Cpl	1.444	Capable
ABF_Roughness	349.325	5.427	300.0	400.0	Cpk	3.03	Excellent

Assessment guide: Excellent ≥ 1.67 , Capable ≥ 1.33 , Marginal ≥ 1.00 , Improvement Required < 1.00

Cpk is used for two-sided specifications. Cpl or Cpu is used for one-sided specifications.

Process Adjustment Recommendations

Recommended Adjustment Priorities

- 1. Press2_Pressure: Increase toward target 10.000 kgf/cm²
- 2. Press2_Temp: Increase toward target 100.000 °C
- 3. CZ_Concentration: Increase toward target 10.000 %
- 4. Anneal_Temp: Increase toward target 200.000 °C
- 5. Cure_Time: Increase toward target 60.000 min

Priority	Parameter	Current_Mean	Normal_Mean	Defect_Mean	Target	Direction
1	Press2_Pressure	9.927	9.96	9.701	10.0	Increase
2	Press2_Temp	99.806	99.877	99.32	100.0	Increase
3	CZ_Concentration	9.999	10.007	9.946	10.0	Increase
4	Anneal_Temp	200.001	200.037	199.761	200.0	Increase
5	Cure_Time	59.95	59.982	59.734	60.0	Increase

Method: Recommendations are ranked using the standardized difference between normal and defect LOTs combined with Random Forest feature importance.

Important: These are directional recommendations derived from synthetic simulation data. They must not be applied directly to production without DOE, engineering review, and process validation.

Quality Parameter Summary

Parameter	count	mean	std	min	25%	50%	75%	max
CZ_Roughness	8000.0	349.931	8.225	318.48	344.33	349.8	355.402	385.25
Total_Thickness	8000.0	32.025	0.262	31.028	31.849	32.021	32.202	33.155
Peel_Strength	8000.0	496.755	22.33	396.95	481.485	496.68	511.82	573.27
ABF_Roughness	8000.0	349.325	5.427	329.67	345.57	349.32	353.09	366.79
Yield	8000.0	97.455	5.729	0.0	98.15	98.53	98.84	100.0